Reproducibility in Bioinformatics

Automated Data Access

What is reproducibility?

The ability to replicate a process.

It may apply to different concepts:

1. Replicating the results of a new discovery
2. Replicating a process by which a discovery was made

When it comes to data analysis we focus on the 2nd definition.

How was the discovery made?

What is "analysis" reproducibility

Understanding what someone did. Compare the two statements:

1. *We have quality controlled the data with the tool called* `trimmomatic` *keeping only data with an average quality of 30.*
2. *We ran* `trimmomatic read1.fq SLIDINGWINDOW:4:30` *for quality control.*

Think about the relative merits of each statement.

If you had to redo the analysis which statement would you prefer?

What should be reproducible?

The result of the analysis need to support the same biological discovery.

It is not about getting the exact same files - we need to obtain results that support the same conclusion.

What if a discovery can only be made with one particular approach?

Is the burden of proof higher or lower for then?

Ingredients of reproducibility

1. Start with the same data

2. Understand what has been done.

How to tell if your analysis is reproducible?

Your analysis is reproducible

if, and only if

you too are able to reproduce it many times over

Reproducibility starts NOW

Reproducibility is not something you add later, at the end of the analysis.

You should **keep re-doing** it all the time. Erase and start over. Is it hard? Fix the hard part.

The only constraint of your work should be the computer run time and not setting it all up and preparing the analysis itself.

If you had to delete all your data and were allowed to keep a single description of what you did how hard would it be to restore the results? That is "reproducibility".

How do we reproduce published results?

We are interested in accessing the data and results for the study titled

- **Genomic surveillance elucidates Ebola virus origin and transmission during the 2014 outbreak**

Published in 2014 in the Science research journal.

Among the many results paper identifies the changes in the virus DNA between the Ebola outbreak of 1976 and 2014.

Typical work strategy

1. Familiarize yourself with the data structure. Often this is best done online.
2. Write command line code that reproducibly obtains the data for accession numbers.

Search NCBI for Ebola

- ebola
- ebola virus
- ebola virus genome 1976

Note the uncertainty there - more precise query may miss the RefSeq entry.

It is never clear when your query is correct!

So it is even more critical to be unambiguous.

Accession numbers

A "well" defined identifier that links to an entry in a database.

Same information may have multiple access points: `10141003`, `AF086833`, `AF086833.2`

Different databases will use different standards.

Your eye can learn to recognize accession numbers. `PRNJA` will be BioProjects.

NCBI specific numbers

Unique ID : a unique (often numerical, `10141003`) identifier as it is entered into the database. Their use is now discouraged but some tools still use them.

Accession Numbers (Loci): an accession number: `AF086833` applies to the complete database record and remains stable even if updates/revisions are made to the record. The data for a loci **may change**.

Version number: an accession number `AF086833.2` with a version number attached to it. These are unique and data under it **never changes**.

We have located accession numbers

`AF086833` or `NC_002549` refer to the same data.

The first is a GenBank sequence, the second is a curated RefSeq sequence.

We can get obtain from the command line with Entrez Direct.

Remember to activate your environment if you get command not found errors:

```
source activate bioinfo
```

Entrez Direct

A command line interface to NCBI. It is a set of tools:

- `esearch` to search databases
- `efetch` to download sequences
- `elink` to connect across databases
- `einfo` to obtain information on databases
- `xtract` to process XML data

See all databases you can search:

```
einfo -dbs
```

Getting data with Entrez Direct

Once you know an accession number like `AF086833` you can fetch it in different formats:

```
efetch -db=nuccore -format=gb -id=AF086833 > AF086833.fa
cat AF086833.fa | head
```

Produces:

```
LOCUS       AF086833                18959 bp    cRNA    lin
DEFINITION  Ebola virus - Mayinga, Zaire, 1976, complete g
ACCESSION   AF086833
VERSION     AF086833.2
KEYWORDS    .
SOURCE      Ebola virus - Mayinga, Zaire, 1976 (EBOV-May)
```

Same accession in a different format

We will cover the details of the format in the future lecture.

```
efetch -db=nuccore -format=fasta -id=AF086833 > AF086833.f
cat AF086833.fa | head
```

Produces

```
>AF086833.2 Ebola virus - Mayinga, Zaire, 1976, complete g
CGGACACACAAAAAGAAAGAAGAATTTTTAGGATCTTTTGTGTGCGAATAACTATGAG
TTTTCCTCTCATTGAAATTTATATCGGAATTTAAATTGAAATTGTTACTGTAATCACA
CAGAGCCACATCACAAAGATAGAGAACAACCTAGGTCTCCGAAGGGAGCAAGGGCATC
TGAAAATCCCTTGTCAACACCTAGGTCTTATCACATCACAAGTTCCACCTCAGACTCT
AACAACCTTAATAGAAACATTATTGTTAAAGGACAGCATTAGTTCACAGTCAAACAAG
```

Searching linked information

Reading the paper (at the very last line) we find that the project accession number is `PRJNA257197`

```
esearch -db nuccore -query PRJNA257197
```

The search alone reports the counts:

```
<ENTREZ_DIRECT>
  <Db>nuccore</Db>
  <WebEnv>NCID_1_118283973_130.14.18.34_9001_1504627953_20
  <QueryKey>1</QueryKey>
  <Count>249</Count>
  <Step>1</Step>
</ENTREZ_DIRECT>
```

The search is database specific

Different databases show different counts:

```
esearch -db protein -query PRJNA257197
```

The search only reports if it has results or not:

```
 <ENTREZ_DIRECT>
        <Db>protein</Db>
        <WebEnv>NCID_1_305224152_130.14.22.215_9001_150463
        <QueryKey>1</QueryKey>
        <Count>2240</Count>
        <Step>1</Step>
</ENTREZ_DIRECT>
```

The search could be asked to return `2240` results.

Obtaining the data

To get the data, pipe the **esearch** into a **efetch**:

```
esearch -db nuccore -query PRJNA257197 | efetch
                                  -format fasta > nucleotides.fa
```

What did we fetch:

```
cat nucleotides.fa | head -4
```

Prints nucleotide sequences:

```
>KR105345.1 Zaire ebolavirus isolate Ebola virus/H.sapiens
ATAATTTTCCTCTCATTGAAATTTATATCGGAATTTAAATTGAAATTGTTACTGTAAT
GTTTCAGAGCCATATCACCAAGATAGAGAACAACCTAGGTCTCCGGAGGGGGCAAGGG
CAGTTGAAAATCCCTTGTCAACATCTAGGCCTTATCACATCACAAGTTCCGCCTTAAA
```

Gets database specific data

To get the data, pipe the search into a fetch:

```
esearch -db protein -query PRJNA257197 | efetch
                                -format fasta > proteins.fa
```

What did we fetch:

```
cat proteins.fa | head -3
```

Prints protein sequences:

```
>AKC37233.1 polymerase [Zaire ebolavirus]
MATQHTQYPDARLSSPIVLDQCDLVTRACGLYSSYSLNPQLRNCKLPKHIYRLKYDVT
LPIDFIVPILLKALSGNGFCPVEPRCQQFLDEIIKYTMQDALFLKYYLKNVGAQEDCV
```

Search different databases

Different databases allow different information to be retrieved. All belonging to the same publication.

```
esearch  -db sra -query PRJNA257197 | efetch
                                -format runinfo > runinfo.csv
```

Now we have a file with lots of columns:

```
cat runinfo.csv | cut -d , -f 1,2,16 | head -3
```

prints:

```
Run,ReleaseDate,LibraryLayout
SRR1972917,2015-04-14 13:59:24,PAIRED
SRR1972918,2015-04-14 13:58:26,PAIRED
```

How to know what information is there?

It is a bit of trial and error. The documentation can be exceedingly obtuse.

```
esearch  -db sra -query PRJNA257197 | efetch
                                    -format summary > summary.xml
```

For example the `runinfo` is a csv file but the `summary` is in XML.

The `summary.xml` is surprisingly large and will download a lot slower.

Why is the NCBI site so confusing?

It is not just NCBI. The vast majority of repositories are like that.

The scope and needs of life scientists exceed the ability of these organizations to act on that need. Most of them are insular are not required to communicate well.

Being a bioinformatician requires the skill to recognize and work around the endless of limitations and problems that might occur.

You have to learn to **deal with it**.

Programmatic access to other data sources

Each data source may have its own way of accessing the data Programmatically

- **NCBI** - Entrez Direct
- **ENSEMBLE** - REST interface
- **UCSC** - mysql database queries

In this course we will use Entrez Direct.

Note: you will need to manually install `mysql` if you want to try out access to UCSC